Prelude

A typical MSc thesis will be structured according to a fairly standard sequence of sections, and one example is shown here. However, it is hard and perhaps even counter-productive to generalise: the goal of this document is *not* to be prescriptive, to show you how you *must* structure your thesis; rather it is to simply to act as a guideline, to show you how you *could* structure your thesis. In particular, each chapter's indicated page-count is important but *not* absolute: the aim is simply to highlight that a clear and concise description is better than a rambling alternative that makes it hard to separate important content and facts from irrelevant trivia.

You can use this document as a LaTeX-based template for your own thesis by simply deleting extraneous sections and content, and adding your own text. If you do that, we recommend using BibTeX to deal with the associated bibliography. You may want to use Overleaf (an online cloud-based collaborative LaTeXplatform).

The words "thesis" and "dissertation" are used interchangeably here. All active hyperlinks in the PDF file are rendered in the same shade of dark blue.

University of BRISTOL

DEPARTMENT OF ENGINEERING MATHEMATICS

Using machine learning to forecast physical activity participation, volunteering and activity patterns in London

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A dissertation submitted to the University of Bristol in accordance with the requirements of the degree of Master of Science in the Faculty of Engineering.

Wednesday 2^{nd} September, 2026

Supervisor: Dr. Dalila O'Grady

Declaration

This dissertation is submitted to the University of Bristol in accordance with the requirements of the degree of MSc in the Faculty of Engineering. It has not been submitted for any other degree or diploma of any examining body. Except where specifically acknowledged, it is all the work of the Author.

Ashrey Ashrey, Sanya Kapoor, Trisha Sharma, Yash Bhooshan Pandloskar, Wednesday 2nd September, 2026

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Abstract

Physical activity participation varies considerably across London boroughs and population groups, contributing to persistent inequalities in health and wellbeing. Understanding how these patterns may change over time is important for informing public health planning, resource allocation and local interventions. This study develops an interpretable framework for analysing and forecasting physical activity participation across London using eight waves of the Sport England Active Lives Adult Survey from 2015–16 to 2022–23. The repeated cross-sectional structure of the survey, changes in variable definitions and geographical identifiers, and the limited number of annual observations present challenges for conventional borough-level time-series forecasting.

A harmonised dataset comprising 135,497 respondents across 32 London boroughs was used to examine relationships between demographic and geographical characteristics and physical activity. Several statistical and machine-learning approaches were considered, including ordinal logistic modelling, LightGBM, CatBoost, K-Modes clustering with classification, and Bayesian hierarchical ordinal modelling. LightGBM provided strong predictive performance among the comparison models, while the ordinal logistic model achieved comparable performance and was selected as the principal propensity model within the forecasting framework because of its simplicity, interpretability and direct representation of the three ordered activity categories. Model evaluation used a final unseen survey wave together with rolling-origin backtesting. The selected forecasting approach achieved a mean absolute error of 2.74 percentage points across the clean forecast origins, compared with 3.44 percentage points for a persistence baseline.

The forecasting framework combines estimated activity propensities with projected demographic composition and a separately estimated behavioural trend, allowing future changes in participation to be distinguished between population-composition effects and behavioural change within demographic groups. Results indicate persistent demographic and geographical inequalities in physical activity across London. Near-term projections suggest broad stability in London-wide activity, resulting from modest behavioural improvement offsetting downward pressure from projected demographic change, while longer-term scenarios to 2031–32 remain sensitive to assumptions about future behaviour. Borough-level differences are also expected to persist, with substantial variation in projected participation across London. Outcomes for which the available survey coverage did not support reliable forecasting were analysed descriptively rather than extrapolated.

Overall, the study demonstrates how repeated cross-sectional survey data can support interpretable population-level forecasting when individual activity propensity is modelled separately from future demographic and behavioural change. The framework provides a practical basis for examining how population composition, behaviour and geographical variation may shape future physical activity participation, while recognising the uncertainty arising from the short historical series and disruption associated with the COVID-19 pandemic.

Project Repository

`https://github.com/ss25231/London-Time-Series-Forecasting-Patterns`

Supporting Technologies

A compulsory section, of at most 1 page

This section should present a detailed summary, in bullet point form, of any third-party resources (e.g., hardware and software components) used during the project. Use of such resources is always perfectly acceptable: the goal of this section is simply to be clear about how and where they are used, so that a clear assessment of your work can result. The content can focus on the project topic itself (rather, for example, than including "I used LaTeX to prepare my dissertation"); an example is as follows:

- I used the *Pandas* and *Seaborn* public-domian Python Libraries.
- I used a parts of the OpenCV computer vision library to capture images from a camera, and for various standard operations (e.g., threshold, edge detection).
- I used Amazon Web Services for remote storage and processing of data. Specifically, I used:
  - Simple Storage Service (S3) for data storage
  - Elastic Compute Cloud (EC2) for provision of virtual machines
  - Elastic Beanstalk for scaling and load management
  - Sagemaker for all the machine learning components of my project.
- I used LaTeX to format my thesis, via the online service *Overleaf*.

Notation and Acronyms

An optional section, of roughly 1 or 2 pages

Any well written document will introduce notation and acronyms before their use, *even if* they are standard in some way: this ensures any reader can understand the resulting self-contained content.

Said introduction can exist within the dissertation itself, wherever that is appropriate. For an acronym, this is typically achieved at the first point of use via "Advanced Encryption Standard (AES)" or similar, noting the capitalisation of relevant letters. However, it can be useful to include an additional, dedicated list at the start of the dissertation; the advantage of doing so is that you cannot mistakenly use an acronym before defining it. A limited example is as follows:

AES	:	Advanced Encryption Standard
DES	:	Data Encryption Standard
	$\vdots$	
$\mathcal{H}(x)$	:	the Hamming weight of x
$\mathbb{F}_q$	:	a finite field with q elements
x_i	:	the i-th bit of some binary sequence x, st. $x_i \in \{0, 1\}$

Acknowledgements

An optional section, of at most 1 page

It is common practice (although totally optional) to acknowledge any third-party advice, contribution or influence you have found useful during your work. Examples include support from friends or family, the input of your Supervisor and/or Advisor, external organisations or persons who have supplied resources of some kind (e.g., funding, advice or time), and so on.

Dave Cliff writes here to say huge thanks to his colleague Dr Dan Page for sharing this LaTeX thesis template, which was originally written by Dan, for Computer Science dissertations. Dave edited Dan's original to better suit the needs of the Data Science MSc: please don't hassle Dan about any of this, but do feel free to contact Dave if you have any questions or comments on it.

Chapter 1

Introduction

1.1 Context and Problem Statement

Physical activity is an important determinant of physical health, mental well-being, and quality of life [16]. However, the participation rate differs from one population group to another and even between regions, which results in health disparities and adds to the burden on public health care systems [16].

London acts as a significant location when it comes to studying the factors affecting the participation rates of people in doing Physical activities due to diverse demographics residing here, having socio-economic disparities and variation in availability of sports facilities as well [14, 2]. It helps understand the obstacles faced by Public Health Care authorities and proves that the benefits gained through exercise are not uniformly distributed among citizens [2, 3].

Understanding how participation may change in the future is therefore relevant to organisations and policymakers when planning interventions, allocating resources and setting local targets. However, population surveys primarily provide information about historical participation rather than direct estimates of future outcomes.

This creates a practical forecasting problem: how can available survey data be used to estimate future participation when the data consist of repeated cross-sectional surveys rather than a panel following the same individuals over time? This research addresses this problem using eight consecutive waves of the Active Lives Adult Survey (2015–16 to 2022–23), covering respondents across 32 London boroughs [14].

1.2 Motivation and Contribution

The significance of this project can be considered from both practical and methodological perspectives. From a practical perspective, borough-level forecasts and demographic-specific estimates can provide local authorities and other organisations with evidence to support target-setting, resource allocation and intervention planning. The analysis also identifies population groups and activity areas where future participation may be more sensitive to demographic or behavioural change.

The project also addresses a gap in the existing literature. Most studies based on the Active Lives Survey have used regression or multilevel model analyses to explore the relationship between demographic factors and physical activity [2, 3]. However, comparatively less attention has been given to forecasting future participation using the survey. This presents an additional methodological challenge because the survey is repeated cross-sectional, while questionnaire structures and variable definitions have changed across waves [14]. Therefore, the project develops a forecasting approach suited to the data's characteristics.

There are only eight annual survey waves, with several affected by the disruption caused by the COVID-19 pandemic [14]. Rather than relying on a complex temporal model to learn borough-specific trends from these limited observations, this project separates the relationship between individual characteristics and activity level from projected changes in population composition and behavioural trends.

The relationship between demographic characteristics and activity level is estimated using the individual-level survey data, while future population composition and behavioural change are projected separately. This approach also makes it possible to distinguish between changes associated with differences in demographic composition and changes in behaviour within demographic groups. This distinction is essential because an increase or decrease in overall participation may not necessarily mean that people's behaviour has changed. It may partly result from changes in population characteristics.

1.3 Central Challenges

The project's design is shaped by three main challenges that have influenced the analytical strategy and the development of the forecasting framework.

Harmonising a survey that changes over time: The Active Lives Survey is a repeated cross-sectional survey rather than a panel survey, which means that a new sample of respondents is collected in each annual wave [14]. The structure of the survey also changed considerably across the eight waves used in this project. For example, the number and availability of recorded variables changed substantially, while some variable definitions, category codes and local-authority identifiers differed between waves. Local-authority identifiers, therefore, required mapping to consistent ONS Government Statistical Service codes [11]. Before the analytical stage, different members of the project team independently identified borough codes that had been incorrectly matched between waves. This highlighted the importance of carefully checking the geographic information before conducting subsequent analysis, as such inconsistencies would silently corrupt every subsequent analytical stage.

Limited temporal information for conventional forecasting: The dataset used contains 32 boroughs across eight annual waves, giving 256 borough-year observations. Several survey waves were also affected by the disruption associated with the COVID-19 pandemic [14]. This provides relatively limited temporal information from which a time-series or machine-learning model could learn reliable long-term patterns. Instead of trying to learn a complex time trend from these few observations, the project therefore separates the relationship between demographic characteristics and activity from the projection of future population composition. This allows the larger individual-level survey sample to be used when estimating the demographic relationship, while population changes are considered separately when producing the forecasts. Given the temporal structure of the data, model evaluation uses a final unseen-wave test, together with rolling-origin back testing, to assess whether the framework can generalise to later survey waves without using information from future periods.

Maintaining interpretability for a non-technical audience: The forecasts are intended to support decision-making by local authorities and organisations around resource allocation and possible interventions. Therefore, interpretability was considered alongside predictive performance and computational efficiency. A more complex model is not necessarily more useful if its results are difficult for the intended audience to understand or justify. The modelling approaches were therefore evaluated based on predictive performance, compatibility with the forecasting objective, computational practicality and interpretability. LightGBM achieved the strongest predictive performance among the comparison models using the three-band activity outcome [7]. However, its performance was close to that of the ordinal logistic model, and the ordinal logistic model was selected as the principal propensity model within the forecasting framework because of its simplicity and interpretability [10]. This consideration is further discussed in Chapter 2, where the candidate modelling approaches are introduced, and in Chapter 4, where their comparison informs the selection of the principal propensity model used within the forecasting framework.

1.4 Aims and Objectives

Based on the challenges discussed above, the overall aim and objectives of this project are as follows:

Aim: To develop an interpretable borough-level framework for analysing and forecasting physical activity and sport participation across London, using harmonised Active Lives Survey data from 2015–16 to 2022–23.

Objective 1: To harmonise the eight Active Lives Survey waves into a consistent dataset by addressing inconsistencies in variable definitions, category coding and geographical coding.

Objective 2: To develop and compare candidate statistical and machine-learning approaches for modelling activity level, and to select the model most suitable for the forecasting framework.

Objective 3: To investigate the extent to which the differences in participation between London boroughs are associated with demographic composition and borough-level differences.

Objective 4: To develop a forecasting framework that combines estimated activity propensities with projected demographic composition and behavioural trends, distinguishing changes associated with population composition from changes associated with behaviour within demographic groups.

Objective 5: To apply the developed framework to overall physical activity and selected demographic and activity measures, using descriptive analysis for outcomes where the available data do not support reliable forecasting, and to present the findings in an accessible form for non-technical stakeholders.

These objectives provide a structured basis for developing and evaluating the borough-level analytical and forecasting framework. Moreover, it also tries to give an explanation of the determinants of the variations in participation among the different London boroughs and how such variations can affect future predictions.

Chapter 2

Background

2.1 Problem Definition

Physical inactivity is one of the major public health issues that exists in London. Sport and physical activity participation levels vary widely across different boroughs and socio-economic classes and play a major role in contributing to health inequalities [4]. The Active Lives Survey is a valuable source of information in this regard; however, some features of the survey should be noted while forecasting. The survey is repeated cross-sectional, meaning that each annual wave contains a new sample of respondents rather than following the same individuals over time. In addition, the structure of the survey, variable definitions and category coding changed across the period covered by this project. During harmonisation of the 2015–16 to 2022–23 waves, for example, one socio-economic classification category was incorporated into an existing category from 2019–20 onwards. Similarly, the mid-level activity category was renamed from 'Insufficiently Active' to 'Fairly Active', although the underlying activity thresholds remained unchanged. These changes required the consistency of variable meanings to be evaluated before combining observations across waves.

Geographical identifiers presented a further challenge. Although the London borough boundaries used in the analysis remained geographically consistent, local-authority codes were not consistent across the survey files. The local-authority codes were therefore mapped using wave-specific metadata to permanent Office for National Statistics Government Statistical Service identifiers, providing a consistent geographical reference across the study period. These issues demonstrate that the forecasting problem cannot be addressed solely through the choice of modelling technique. Reliable forecasting also depends on establishing comparable observations across survey waves. Harmonisation and validation were therefore treated as an integral part of the analytical process, providing the foundation for the modelling and forecasting procedures developed in the study.

2.2 Domain Background: Determinants of Participation

Some of the demographic and geographic variables associated with physical activity engagement include: age, gender, socio-economic status, geographic location, and characteristics of the neighbourhood [2, 3, 5]. These findings provide a basis for considering demographic and geographical characteristics when modelling activity levels. The present study considers age, gender, ethnicity, socio-economic classification, education, disability status, body mass index group, and borough identity as potential predictors of activity level. These characteristics were therefore examined in the exploratory and predictive analyses, with their relationships with activity level considered across the alternative modelling approaches.

Existing research using the Active Lives Survey has largely focused on describing participation patterns and examining associations between demographic characteristics and physical activity. Regression and multilevel approaches are useful for understanding these relationships, but identifying associations alone does not provide a mechanism for forecasting future participation. The present study, therefore, extends this perspective by incorporating the relationship between individual characteristics and activity level into a borough-level forecasting framework, rather than treating historical participation rates as the sole basis for future projections.

2.3 Forecasting Physical Activity

Forecasting physical activity presents additional challenges when the available data are repeated cross-sectional rather than longitudinal. Traditional time series models like the ARIMA-based models depend on past observations that provide temporal information in sufficient amounts to help predict future values [1]. Some of the newer forecasting methods, like Prophet, Temporal Fusion Transformers, and N-BEATS, have the capability of modelling complex temporal patterns [15, 9, 12]. However, the present study contains only eight annual survey waves at the borough level. At the same time, the COVID-19 disruption introduces an unusual period within the observed series, making direct extrapolation of borough-level participation rates less reliable.

Conventional time-series approaches such as ARIMA and Prophet provide established methods for temporal forecasting. However, the limited number of annual observations in the present study limited their suitability for the main forecasting analysis. Machine-learning methods have also increasingly been used to model physical activity and related behavioural outcomes. The Active Lives data provide a large individual-level sample containing demographic and geographical characteristics that can be used to estimate differences in activity propensity. Methods such as LightGBM can capture non-linear relationships and interactions between predictors and provide a flexible framework for modelling structured tabular data [7].

However, predictive flexibility does not overcome the survey's limited temporal information. The forecasting problem, therefore, requires a framework that can make use of the much larger individual-level sample without relying on a long borough-level time series. The approach adopted in this study consequently separates the estimation of activity propensity from the projection of future population characteristics. Individual demographic characteristics are used to estimate activity level, while projected changes in population composition and behavioural trends are incorporated separately when generating future borough-level estimates.

2.4 Modelling Approaches Considered

Building on the demographic and geographical determinants identified above and the limitations of direct temporal forecasting, several modelling approaches were considered to assess different ways of representing physical activity participation. Machine-learning approaches, including **LightGBM and CatBoost**, were considered as comparison models because of their ability to capture non-linear relationships and interactions between predictors [7, 13]. K-Modes clustering and classification were also considered as an alternative in order to determine if respondents with similar demographic and geographic characteristics can be segmented into different population groups. K-Modes is specifically suited for categorical data analysis and, therefore, was ideal for the exploratory study at hand [6]. In the K-Modes framework, Random Forest and Gradient Boosting models were used in order to determine the effectiveness of classification in the segmentation process.

The Bayesian hierarchical ordinal model was also another model that was considered in light of the ordinal nature of the activity outcome and different boroughs for each respondent. Ordinal regression provides one with the means of analysing ordered categories [10], while the hierarchical structure takes into account geographical variations through borough effects. This approach provided a way to model individual-level relationships while accounting for variation between London boroughs and was used for methodological comparison rather than as the final forecasting model.

For the forecasting framework, particular consideration was given to the ordinal logistic model because the outcome consists of three ordered activity categories: inactive, fairly active and active. This provides a direct and interpretable way of estimating the probability of an individual belonging to each activity category based on their demographic and geographical characteristics. LightGBM achieved the strongest predictive performance among the comparison models using the three-band outcome, while the performance of the ordinal logistic model was close to that of LightGBM. **The ordinal logistic model was therefore selected as the principal propensity model within the forecasting framework** because of its simplicity and interpretability, while providing predictive performance suitable for the modelling stage.

2.5 Modelling and Forecasting Framework

The candidate approaches provided different ways of modelling physical activity participation, but the final framework required an approach that could be integrated with future population projections. The study therefore adopts an individual-level propensity modelling approach rather than forecasting borough participation rates directly from the short sequence of annual observations. The selected framework estimates activity propensity based on individual demographic and geographical characteristics, using the harmonised Active Lives data. The activity outcome is represented using three ordered activity bands: inactive, fairly active and active. The evaluation therefore considers both performance on a final unseen survey wave and rolling-origin backtesting across earlier forecast origins.

Although LightGBM achieved the strongest predictive performance among the comparison models using the three-band outcome, its performance was close to that of the ordinal logistic model. The ordinal logistic model was therefore selected as the principal propensity model within the forecasting framework because of its simplicity and interpretability. Model selection considered predictive performance, compatibility with the three-band forecasting target, computational practicality, compatibility with the wider forecasting pipeline and interpretability.

The estimated propensities are subsequently combined with projected population composition and estimated behavioural trends to generate future borough-level participation estimates. For the final propensity modelling framework, survey weights were incorporated where appropriate to account for the survey design when estimating participation patterns. Forecast performance was also assessed against persistence and drift baselines, representing continuation of the latest observed rate and continuation of the previous trend, respectively.

The framework was applied to the main analytical areas identified in the study, with the extent of forecasting determined by the temporal coverage, consistency and predictive evidence available for each outcome. Outcomes for which these conditions were not satisfied were treated descriptively rather than extrapolated.

2.6 Population Composition and Geographical Variation

The forecasting framework adopts a composition-place perspective to distinguish between changes in participation that arise from population characteristics and those associated with differences between geographical areas. This distinction is important because borough-level participation rates can change even when the underlying behaviour of individual population groups remains similar, simply because the population composition changes over time. The general distinction between differences arising from population composition and differences in group-specific rates follows the demographic decomposition approach introduced by Kitagawa [8]. The individual-level modelling component estimates activity propensities from demographic and geographical characteristics. These estimates can then be combined with projected population composition to determine how changes in population characteristics may affect future participation. Behavioural trends are considered separately so that changes in predicted participation are not attributed entirely to demographic change.

This perspective also allows the analysis to examine whether observed differences between London boroughs remain after accounting for differences in population characteristics. The exploratory analysis and Bayesian hierarchical modelling allow the study to assess whether borough-level variation persists after accounting for observed demographic characteristics, with the hierarchical model capturing residual geographical variation via a borough-level random effect. This analysis indicated that borough-level differences were not fully accounted for by the observed demographic characteristics, supporting the consideration of geographical variation as a distinct component of the framework. The framework provides an interpretable basis for understanding whether differences in participation are more closely associated with demographic composition or with differences in activity among comparable demographic groups. At the same time, the hierarchical analysis assesses residual borough-level variation separately.

2.7 Research Gap and Related Work

Enough research evidence has provided a solid foundation for the relation between physical activities and demographics, socioeconomic status, and geography [5, 2, 3]. However, the literature considered in this study largely focuses on describing participation patterns or estimating associations between individual characteristics and physical activity, rather than developing a borough-level forecasting framework from repeated cross-sectional survey data. The methodological gap addressed by this study, therefore, concerns how individual-level relationships can be combined with projected population composition and behavioural change when only a short sequence of annual survey observations is available.

For instance, the recent machine learning research [4] reveals the possibility of predicting physical activity recommendations based on the individual survey characteristics. However, the study focuses on current individual-level prediction rather than future borough-level forecasting, leaving scope for an approach that combines individual-level relationships with population and geographical change. The Active Lives Survey provides an opportunity to extend this work. However, its repeated cross-sectional structure, changing survey characteristics and limited number of annual observations create challenges for conventional borough-level forecasting. This motivates an approach that uses individual-level information while accounting for changes in population composition and geographical variation.

This study addresses this gap through an interpretable forecasting framework that estimates individual activity propensities and combines them with projected population composition and a separately estimated behavioural trend. The framework also allows differences between boroughs to be considered alongside differences in population characteristics, providing a basis for examining how population composition and behavioural change may influence future participation.

Chapter 3 presents the data, harmonisation procedures, exploratory analysis, candidate models and forecasting methodology used to implement the framework. Chapter 4 evaluates the candidate models and presents the resulting analysis and forecasts.

Chapter 3

Methodology

3.1 Overall Project Workflow

This project follows a five-stage modelling and forecasting workflow, designed to transform the Active Lives Survey data into borough and London level forecasts. The process begins with data preparation and quality checks, followed by feature engineering to create a consistent modelling dataset. Different modelling approaches are then developed to examine activity participation from different perspectives. The final stage combines the modelling results with projected demographic trends to produce forecasts and scenario analysis.

Stage 1 – Data Preparation and Harmonisation: The eight annual survey waves were combined into a single dataset. Variables were standardised across years, borough identifiers were harmonised using permanent ONS GSS codes, and missing or inconsistent values were addressed.

Stage 2 – Exploratory Data Analysis: The cleaned data were examined to identify missingness, unusual values, changes in survey coverage and other potential data-quality issues before modelling.

Stage 3 – Feature Engineering: The Activity Level outcome was derived and demographic variables were consolidated into a model-ready dataset.

Stage 4 – Machine Learning Modelling: Four separate modelling approaches were developed: LightGBM, CatBoost, clustering-based models, and a Bayesian hierarchical ordinal model.

Stage 5 – Forecasting and Scenario Analysis: The selected forecasting model was combined with projected population composition and behavioural trends to estimate future activity levels and separate changes associated with population composition from changes in behaviour.

A time-based validation strategy was used, training models on earlier survey waves and testing them on later waves to avoid using future information in the predictions.

3.2 Dataset Description

The analysis uses data from Sport England's Active Lives Adult Survey, a nationally representative survey of adults aged 16 and over. The survey collects information on physical activity, demographic characteristics, health, education, volunteering and other lifestyle factors. As the survey uses a repeated cross-sectional design, different respondents are surveyed in each wave rather than following the same individuals over time.

Eight survey waves from 2015–16 to 2022–23 were used. The national data were filtered to the 32 London boroughs, excluding the City of London, resulting in 135,497 respondents. The number of variables varied between survey waves because additional question blocks were introduced over time. Therefore, the data required harmonisation before the waves could be combined into a consistent dataset.

Borough information was standardised using ONS geographical codes, with the different local-authority variables used across survey years mapped to a common `LA_code`. Variables that changed names or labels between waves were also aligned using the survey metadata. This included changes to demographic and activity-related variables, while keeping the underlying categories and definitions consistent.

Property	Value
Data source	Sport England Active Lives Adult Survey (UKDS Ref. 9288)
Survey period	2015–16 to 2022–23
Survey waves	8
Study area	32 London boroughs
Total respondents	135,497
Raw variables (per wave)	∼10,700
Merged dataset variables	610

Table 3.1: Dataset Summary

3.3 Data Preparation, Quality Checks and Feature Engineering

The raw survey waves could not be used directly for modelling because variables, coding and question availability changed across survey years. A structured preparation process was therefore applied before developing the predictive models. The main steps were variable harmonisation, missing-value treatment, consistency checks and construction of the final modelling features.

Data quality and harmonisation. Each survey wave was checked for merge integrity, borough coverage, missing values, sentinel codes and consistency of variable definitions across years. Borough identifiers differed between survey waves, so the relevant local-authority variables were mapped to a common ONS GSS borough code (`LA_code`). This ensured that the same 32 London boroughs could be identified consistently across all eight waves.

The Active Lives Survey also uses negative values to represent different types of missing or non-applicable responses. These were treated as missing rather than numerical observations. The SPSS import process converted the recognised sentinel values to NaN. One additional invalid value identified in the team-sport duration variable was also replaced with NaN.

Missingness was assessed rather than automatically imputed. High missing rates were generally explained by variables being introduced in later survey waves or by survey routing, where questions were only presented to particular respondents. For example, free-time barrier variables were only available in 2022–23, while volunteering and indoor/outdoor measures were introduced after the first wave. These variables were therefore used only where their availability supported the relevant analysis.

Cross-year checks were also carried out for key demographic and activity variables. Where category labels changed but the underlying coding remained the same, the common coding was retained. Where categories genuinely differed, they were recoded to maintain consistency; for example, the additional NSSEC5 category used in earlier waves was mapped to the later equivalent category.

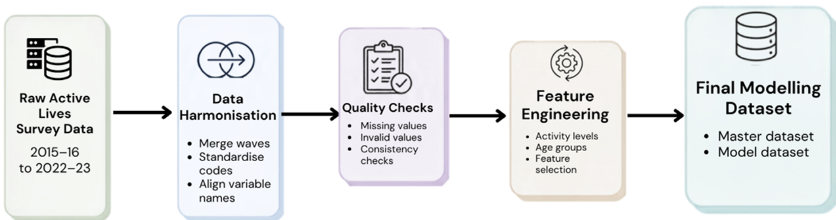

Figure 3.1: Data preprocessing workflow showing dataset harmonisation, data quality checks, feature engineering, and creation of the final modelling datasets.

Feature engineering. After these checks, the cleaned 610-column master dataset was used to create smaller model-specific datasets. The main outcome, `Activity_Level`, was used in an ordered three-category variable: 0:Inactive, 1:Fairly Active, and 2:Active. The demographic predictors were consolidated into consistent categories covering characteristics such as age, gender, ethnicity, disability, socio-economic group and education. Variables were selected according to their relevance to the four analytical pillars and the requirements of each modelling approach.

The resulting features provided a consistent basis for model development while retaining the demographic, geographical and behavioural information required for forecasting and scenario analysis.

3.4 Exploratory Data Analysis

Exploratory analysis was carried out to understand the main patterns in physical activity across London and to identify potential issues that could affect modelling. Activity levels varied considerably across boroughs, with clear differences also observed across demographic groups. In particular, age, socio-economic group and disability status showed noticeable associations with activity, while gender differences were comparatively smaller. These patterns provided an initial indication that demographic composition would be important when modelling participation.

Temporal analysis also showed changes in activity across the survey period, including a clear disruption during the COVID-19 period. This supported the use of time-based validation and forecasting approaches, rather than treating all survey waves as interchangeable observations.

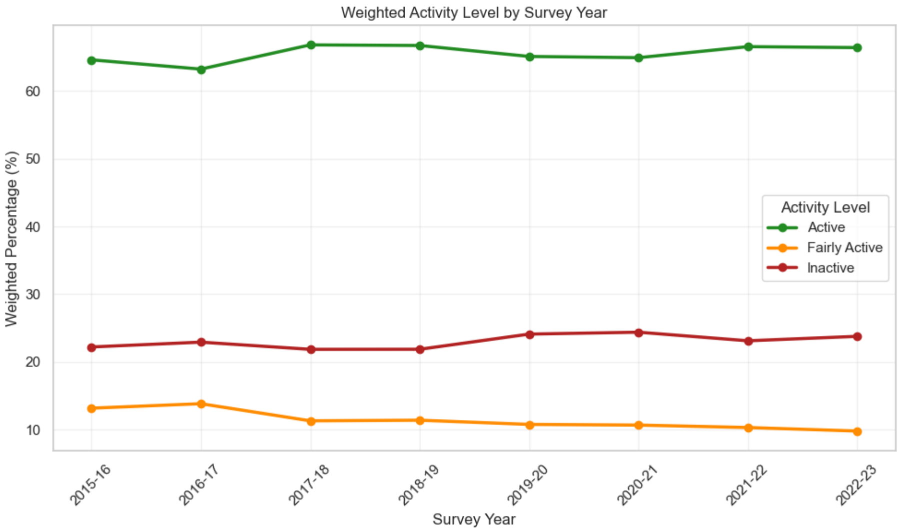

Figure 3.2: Population-weighted distribution of activity levels across survey years

Borough-level analysis further showed that differences in activity were not explained entirely by demographic composition. Some variation remained between boroughs after accounting for observed characteristics, supporting the inclusion of borough-level effects in the modelling framework.

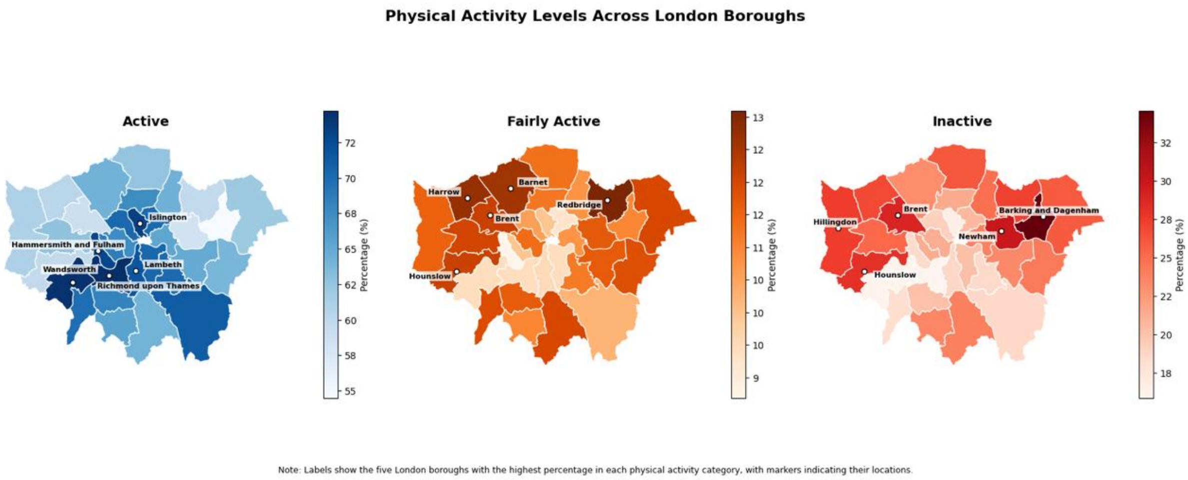

Figure 3.3: Geographical variation in population-weighted activity levels across London's boroughs.

The EDA also identified differences in the availability and consistency of certain survey measures across waves. Variables such as indoor/outdoor activity, volunteering and free-time constraints were available for fewer years than the core activity measures. Consequently, these measures were used selectively in later analysis rather than being treated as consistent long-term forecasting variables.

Overall, the EDA established three important features of the data: strong demographic differences, meaningful temporal variation, and remaining borough-level variation. These findings informed the choice of modelling approaches and the subsequent forecasting framework.

3.5 Modelling Framework

Four modelling approaches were developed to examine physical activity from complementary statistical and machine-learning perspectives: LightGBM, CatBoost, K-Modes clustering with classification, and a Bayesian hierarchical ordinal model. All the models used the same prepared survey data but their outcome definitions and modelling objectives differed. LightGBM and CatBoost were used for supervised prediction, the clustering approach explored respondent segments, and the Bayesian model directly represented the ordered Activity Level outcome and borough-level variation.

3.5.1 Prediction Targets and Notation

Let X_i represent the predictor variables for respondent i, and let Y_i denote their physical activity outcome. For the binary prediction task, the outcome was defined as whether the respondent met the 150-minute physical activity guideline:

$$Y_i = \begin{cases} 1, & \text{if respondent meets the guideline} \\ 0, & \text{otherwise.} \end{cases}$$

The corresponding predicted probability is denoted by

$$p_i = P(Y_i = 1 \mid X_i).$$

This probability represents the model's estimated likelihood that respondent i meets the guideline.

For the ordinal modelling task, physical activity was represented using the three-level `Activity_Level` variable:

$$Y_i \in \{0, 1, 2\},$$

where 0 = Inactive, 1 = Fairly Active and 2 = Active. This preserves the natural ordering of the activity categories rather than treating them as unrelated classes.

3.5.2 Survey Weighting

The Active Lives Survey uses survey weights to account for the sampling design and improve the representativeness of estimates. Where supported by the modelling method, these weights were incorporated during model training. For an observation i with survey weight w_i, a weighted loss can be expressed generally as

$$\mathcal{L} = \sum_{i=1}^{n} w_i \updownarrow(y_i, \widehat{y}_i),$$

where $\updownarrow$ represents the loss for an individual observation. Consequently, observations with larger survey weights contribute more to the model's objective. This was particularly important when developing population-level predictions from the survey data.

3.5.3 Model Specifications

LightGBM was used as one of the candidate supervised models for the forecasting task. It is a gradient-boosting algorithm that builds a sequence of decision trees, with each tree improving the errors made by the previous trees. The prediction can be represented as,

$$\widehat{y}_i = \sum_{k=1}^{K} f_k(X_i),$$

where f_k represents an individual decision tree and K is the total number of trees. This approach is suitable for the survey data because it can capture non-linear relationships and interactions between demographic, temporal and activity-related predictors, and it served as a strong comparison against the ordinal logistic model used in the forecasting framework.

CatBoost was used as a second gradient-boosting approach for comparison. It is designed to handle categorical predictors effectively and can model non-linear relationships without requiring the same level of manual transformation as conventional regression models. SHAP values were additionally used with CatBoost to examine the contribution of individual predictors to model predictions.

K-Modes Clustering and Classification was used to group respondents according to similarities in their categorical demographic and activity characteristics. Unlike supervised prediction, clustering does not require a predefined outcome. The resulting clusters were then used as the target for a supervised classifier to assess how reliably cluster membership could be predicted from respondent characteristics. This provided a population-segmentation perspective rather than a direct forecasting model.

Bayesian Hierarchical Ordinal Model directly treated `Activity_Level` as an ordered outcome and incorporated borough-level variation through a hierarchical structure. The linear predictor was defined as

$$\eta_i = \beta^T X_i + u_{b[i]},$$

where β represents the effects of the observed predictors and $u_{b[i]}$ represents the random effect for the borough containing respondent i. The cumulative probability of belonging to category k was modelled as

$$P(Y_i \leq k) = \text{logit}^{-1}(\theta_k - \eta_i),$$

where θ_k represents the ordered thresholds between activity categories. This allows the model to account for both individual characteristics and differences between boroughs while retaining the ordinal structure of the outcome.

3.5.4 Model Evaluation

Model performance was assessed using metrics appropriate to the prediction task. Accuracy measures the proportion of correctly classified observations, while Precision and Recall measure the reliability and completeness of positive predictions. Their harmonic mean is summarised by the F1-score:

$$F1 = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$

For probabilistic predictions, Log Loss measures how close the predicted probabilities are to the observed outcomes:

$$\text{LogLoss} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right].$$

Lower Log Loss indicates better-calibrated probabilistic predictions. AUC measures the model's ability to distinguish between respondents who meet and do not meet the activity threshold across different classification thresholds, with a value of 0.5 representing random discrimination and higher values indicating better discrimination.

Where applicable, MAE and RMSE were also used for forecasting error:

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \widehat{y}_i|$$

and

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \widehat{y}_i)^2}.$$

MAE gives the average absolute prediction error, while RMSE places greater emphasis on larger errors. These measures were used where the model produced continuous forecasts, while classification metrics were retained for models producing categorical or probabilistic outcomes.

3.6 Model Validation

Model validation was designed to assess how well the models generalise to observations that were not used during training. Since the data are collected across consecutive survey waves, a time-based validation strategy was used rather than a random train–test split. This ensures that information from later years is never used to predict an earlier year.

3.6.1 Temporal Holdout

The final survey wave, 2022–23, was retained as an unseen test set. Models were developed using the preceding survey waves and evaluated on this final wave. This provides a realistic assessment of how the modelling framework performs when applied to a future observation.

3.6.2 Rolling-Origin Backtesting

A rolling-origin approach was additionally used to test whether forecasting performance was consistent across different points in time. At each forecast origin τ, the model was trained only on the waves available up to that point and used to predict the following wave:

$$\text{Train on waves } 1, \dots, \tau \rightarrow \text{Forecast wave } \tau + 1$$

This process was repeated for the available clean forecast origins ($\tau = 3, 6, 7$). Origins that would forecast a pandemic-disrupted wave were excluded from the headline assessment, as these unusual conditions could distort the comparison.

Forecast origin	Training data	Forecast target	Used in headline results
Wave 3	Waves 1–3	Wave 4	yes
Wave 6	Waves 1–6	Wave 7	yes
Wave 7	Waves 1–7	Wave 8	yes
Pandemic-target origins	Available historical waves	Pandemic wave	yes

Table 3.2: Rolling-origin validation design

3.6.3 Forecasting Baselines

Simple forecasting baselines were included to determine whether the modelling framework provided improvement over straightforward alternatives. Two benchmarks were considered:

- Persistence: each borough is assumed to remain at its most recently observed activity rate.
- Drift: each borough is assumed to continue following its previously observed trend.

Model performance was expressed relative to these benchmarks using a skill score:

$$\text{Skill} = 1 - \frac{\text{MAE}_{\text{model}}}{\text{MAE}_{\text{benchmark}}}$$

A positive skill score indicates that the model has lower forecasting error than the benchmark.

3.7 Forecasting Framework

The forecasting framework combines the estimated activity propensities with projected changes in the demographic composition of each borough. This approach was chosen because the dataset contains many individual-level observations but only eight annual survey waves, making a conventional borough-level time-series model unsuitable. Instead, the framework separates changes in participation into demographic composition and behavioural change.

3.7.1 Forecasting equation

For borough b at time t, the predicted activity rate is defined as:

$$\widehat{r}_{b,t} = \sum_c s_{b,c,t} p_{b,c} + \delta_t$$

where $s_{b,c,t}$ is the projected population share of demographic cell c in borough b, $p_{b,c}$ is the estimated probability of being active for that demographic cell, and δ_t represents the overall behavioural trend.

This can be understood as who lives in the borough × how likely that group is to be active + changes in behaviour over time. The propensity model is treated as time-free, meaning that it does not itself predict future time periods. Instead, future change is introduced through the projected population composition and behavioural trend.

3.7.2 Population projection

Demographic composition was projected using the historical survey data for each borough. For each demographic cell, the historical population share was modelled using a linear trend on the logit scale. Borough-specific trends were then shrunk towards the corresponding London-wide trend:

$$\widetilde{m}_{b,c} = \lambda m_{b,c} + (1-\lambda) m_c^{London}, \qquad \lambda = 0.5$$

The projected shares were then transformed back to proportions and normalised so that the demographic shares within each borough sum to one. This shrinkage reduces the risk of unrealistic long-term projections caused by extrapolating from only eight observations.

The projection therefore focuses on demographic characteristics that can be projected reliably from the available population structure. The forecasting framework does not assume that every survey variable can be projected into the future.

3.7.3 Behavioural effect

The behaviour effect represents changes in activity that cannot be explained by changes in the demographic composition of the population. It is estimated from the difference between the observed London activity rate and the rate predicted from the observed demographic composition:

$$\delta_t = r_t^{obs} - \sum_c s_{c,t} p_c$$

A trend is then fitted to this residual using the non-pandemic waves. The COVID-affected waves are treated separately so that the unusual disruption does not determine the long-term behavioural trend. In the analysis, the estimated non-pandemic behavioural change was approximately +0.23 percentage points per year, while the estimated pandemic effect was approximately −2.07 percentage points.

3.7.4 Decomposing forecast changes

The framework allows the change in activity between two periods to be separated into two components:

$$\Delta r_b = \underbrace{\sum_c \Delta s_{b,c} p_{b,c}}_{\text{composition effect}} + \underbrace{\Delta\delta}_{\text{behaviour effect}}$$

The composition effect therefore measures the change resulting from the population becoming demographically different, while the behaviour effect measures the change in activity among people of the same demographic type.

This distinction is important for interpretation because the two effects imply different policy responses. For example, an increase in the proportion of older residents may reduce the overall activity rate through composition, whereas increased participation among existing demographic groups represents behavioural improvement.

3.7.5 Forecast scenarios and aggregation

Future forecasts are produced by applying the projected demographic shares and behavioural trend to each borough. Borough-level estimates are then aggregated to obtain London-wide forecasts using the corresponding population shares.

Because behavioural change is the least certain component over a ten-year horizon, forecasts are interpreted as scenarios rather than precise predictions. Where the available data are insufficient to support a reliable future projection, the analysis reports descriptive findings instead. For example, free-time constraints are available only in the 2022–23 wave, while volunteering and indoor/outdoor measures have limitations in their coverage; these are therefore not treated as equivalent to the main physical-activity forecast.

Chapter 4

Results and Critical Evaluation

4.1 Experimental Results and Validation

The modelling framework was evaluated using both the final unseen survey wave and rolling-origin backtesting. The final 2022–23 wave was kept separate from model development and provided the main test of how well the models generalised to a previously unseen period. In addition, rolling-origin validation was used to assess whether the forecasting approach remained reasonably consistent when predicting different survey waves rather than relying on a single holdout period.

The results show that the supervised models were able to identify meaningful patterns in physical activity from demographic and borough-level characteristics. However, the models differed in their objectives and outcome definitions, so their results cannot be interpreted as a single ranking based on one metric. LightGBM and the Bayesian model predict the three ordered activity levels, whereas CatBoost and the clustering-based approaches use binary or cluster-based outcomes. The comparison therefore considers predictive performance alongside the suitability of each approach for the project's forecasting objective.

A simple forecasting baseline was also used as a reference point. The purpose of the baseline was to establish whether the modelling approach provided useful predictive information beyond what could be achieved by simply carrying forward recent activity levels or continuing an existing trend. This provides a more meaningful assessment of forecasting performance than considering model accuracy alone.

For classification models, performance was assessed using measures including AUC, accuracy, F1-score and Log Loss where available. AUC indicates how well the model distinguishes between different outcome groups, while accuracy and F1-score describe classification performance. Log Loss provides an assessment of the quality of the predicted probabilities. Where continuous forecasting errors were available, MAE and RMSE were also considered, with RMSE giving greater weight to larger errors.

4.2 Model Results and Critical Evaluation

The modelling approaches produced different strengths and limitations because they used different target formulations and modelling assumptions. Therefore, models were compared using predictive performance alongside their suitability for the project's forecasting objective.

Model	**Target**	**AUC**	**Logloss**	**Accuracy**	**Marco F1**	**Runtime**
Light GBM	3-band ordinal	0.707	0.794	67.4%*	—	15 min 50 sec
Catboost	Binary	0.708	0.588	68.9%	—	—
Random Forest – clustering	Binary	—	—	54%	0.428	0 min 14 sec
Gradient Boosting – clustering	Binary	—	—	67.9%	0.363	2 min 16 sec
Bayesian Hierarchical Ordinal*	3-band ordinal	0.68	0.82	65%	0.57	∼96 min

Table 4.1: Model-specific performance metrics

*Weighted accuracy. The Bayesian results are based on the available model run and are interpreted cautiously because the model was considerably more computationally demanding and used a different modelling setup.

4.2.1 Predictive performance

Table 4.1 summarises the available predictive performance measures. LightGBM achieved an AUC of 0.707 and weighted accuracy of 67.4%, while CatBoost achieved an AUC of 0.708 and accuracy of 68.9%. Although CatBoost has slightly higher values, the difference should not be interpreted as a direct ranking because LightGBM predicts the three ordered activity categories, whereas CatBoost uses a binary outcome.

The clustering approaches produced mixed results. Random Forest clustering achieved 54.0% accuracy and a Macro F1 of 0.428, while Gradient Boosting clustering achieved 67.9% accuracy and a Macro F1 of 0.363. The Bayesian hierarchical ordinal model achieved an AUC of 0.680, accuracy of 65.0%, and Macro F1 of 0.570. These results highlight that the approaches serve different modelling purposes and are therefore not directly interchangeable.

4.2.2 Forecasting performance

The selected ordinal logistic propensity model, combined with the demographic-projection framework, was additionally evaluated using rolling-origin backtesting to assess its ability to forecast future activity rates. Across the clean forecast origins, it achieved a mean MAE of 2.74 percentage points, compared with 3.44 percentage points for the naïve baseline. The model achieved mean skill scores of +0.20 against the naïve baseline and +0.31 against the drift benchmark, indicating improved forecasting performance over these simple alternatives.

The CatBoost forecasting implementation produced a final 2022–23 borough-level MAE of 3.28 percentage points and RMSE of 4.03 percentage points across 32 boroughs. However, these results should not be treated as a direct error comparison: the ordinal logistic forecasting value is averaged across rolling forecast origins, whereas the CatBoost values represent the final forecast period.

4.2.3 Selection of the Ordinal Logistic Model

The ordinal logistic model was selected as the propensity model for the forecasting framework because it provided the best overall fit with the project's forecasting objective. Most importantly, the framework requires the three ordered activity categories—Inactive, Fairly Active and Active—and the ordinal (cumulative-logit) formulation represents this ordered structure directly. Its predictive performance was effectively equal to that of the gradient-boosting models (AUC 0.704 against 0.708 for LightGBM), so it was preferred on the grounds of equal accuracy with greater simplicity, interpretability, and more stable behaviour when projected populations are pushed through it.

LightGBM and CatBoost remained important comparison models: their similar AUC demonstrates that the predictive relationships are not dependent on a single algorithm, which strengthens confidence in the selected model rather than diminishing it.

The selection of the ordinal logistic model therefore reflects its overall suitability for the forecasting framework, while the close agreement with LightGBM and CatBoost confirms that the predictive relationships are robust across modelling approaches.

Model	**Strength**	**Limitation**	**Role**
Ordinal Logistic Model	Strongest performance and three-band formulation		Main forecasting model
LightGBM	Strong performance and three-band formulation	Lower interpretability	Best Comparison model
CatBoost	Competitive binary prediction	Binary target	Comparison model
Random Forest clustering	Simple segmentation	Lower predictive performance	Exploratory
Gradient Boosting clustering	Captures non-linear patterns	Lower Macro F1	Exploratory
Bayesian hierarchical ordinal	Explicit ordinal structure	Higher computational cost	Methodological comparison

Table 4.2: Model characteristics and analytical roles

4.3 Physical Activity and Volunteering Trends Over the Next Ten Years

This section reports the forecast of all the four analytical pillars: overall participation forecasts, demographic-specific insights, indoor versus outdoor activity, and volunteering in sport.

A note on forecasting versus description:

Forecasting was applied only where the data provided sufficient evidence for reliable projection: the question had to be measured consistently across survey years, have adequate sample coverage, and show sufficient predictive structure from demographic variables. Overall activity, activity types, club membership and key demographic groups met all three criteria and were therefore forecast, with borough-level projections and quantified uncertainty.

Where these conditions were not met, results are presented descriptively rather than as forecasts, so that no figure implies more confidence than the data can support. Free time was asked only in the final wave, so there is no trend to project and it can only be described as a current snapshot. Volunteering was measured inconsistently, as its question base changed twice during the series, and demographics explain only around 15% of the variation between boroughs-too weak a signal to drive a projection, so it is reported at London and demographic level rather than forecast. Indoor and outdoor activity was asked of just ∼30% of respondents, a rotating subsample that leaves the estimates with considerably greater uncertainty, so it is described with only one cautious London-level projection. Health was unavailable in the early survey years, leaving only five years of data, two of them affected by COVID and excluded as atypical; the remaining three clean years are not sufficient to establish a reliable trend, so despite being the strongest single correlate of activity, it too is reported descriptively.

This distinction ensures that forecasts are supported by the available evidence rather than presenting uncertain estimates as precise future outcomes

4.3.1 P1 - Overall Participation Forecasts

Physical activity across London as a whole

The model projects broad stability in London's overall activity levels in the near term. By 2025–26, around 64.9% of adults are projected to be Active, while 26.0% are Inactive and 9.1% are Fairly Active. Although the Active share changes little, the Fairly Active group has declined steadily over the observed period, suggesting a gradual shift away from the middle activity band.

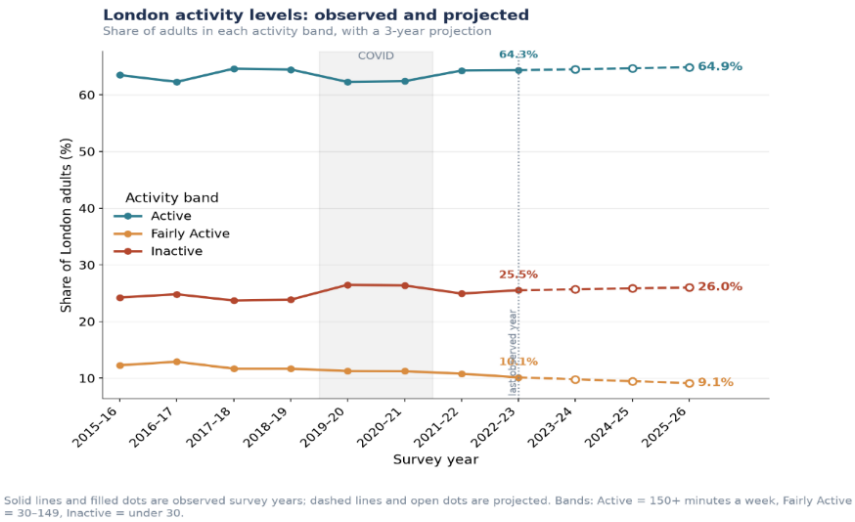

Figure 4.1: London-wide activity bands, observed 2015–16 to 2022–23 and projected to 2025–26

This decomposition is the central analytical result behind the 'stable' headline. A conventional forecast would report the number as roughly flat and stop there. Decomposing it shows that the stability is a coincidence of two genuine trends cancelling, not a settled equilibrium. If the behavioural improvement stalls, the demographic headwind takes over. The stability is more fragile than the flat line suggests.

For the longer-term outlook, three scenarios are presented rather than a single point forecast. By 2031–32, the Active share ranges from approximately 61.0% under a trend-reversal scenario to 65.4% if

Component	Effect by 2025–26	Interpretation
Composition	−0.37 points	London's ageing population pulls activity down
Behaviour	+0.69 points	People of a given type are becoming more active
Net	+0.32 points	Apparent stability produced by two real trends cancelling

Table 4.3: Decomposition of the near-term London-wide forecast into composition and behaviour effects.

recent gains continue, with a population-change-only scenario at around 63.1%. This range highlights that the long-term direction of physical activity is sensitive to the assumptions made about future behaviour.

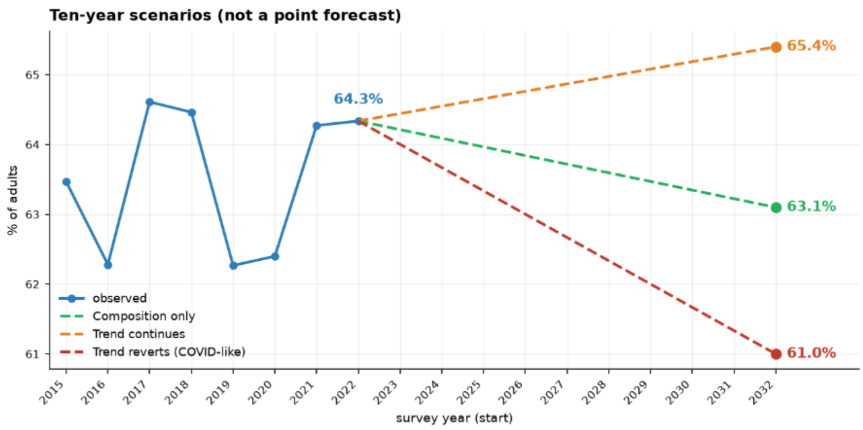

Figure 4.2: Ten-year scenario range for London-wide activity, 2022–2032

Likelihood of participating in specific activities

Ten-year scenario and three-year scenario ranges were produced for seven individual activities and three club-membership measures. Among them the top two activities are: Walking and Active Travel.

Activity / Club Type	Today	If Gains Continue	Population Change Only	If Gains Reverse
Walking	75.7%	71.4%	74.6%	71.5%
Active travel	60.2%	49.1%	58.6%	45.1%
Fitness activities	35.4%	31.8%	34.5%	29.5%
Cycling	19.7%	20.1%	18.9%	17.2%
Team sports	8.2%	6.5%	7.7%	4.9%
Dance	8.9%	5.2%	8.7%	7.0%
Racket sports	6.1%	5.1%	6.0%	4.0%
Club membership (any sport)*	16.3%	20.1%	15.6%	16.5%
Fitness club / gym	10.0%	9.4%	9.5%	9.5%
Team-sport club	4.7%	7.1%	4.3%	4.7%

Table 4.4: Ten-year scenario range by activity/club type (% of London adults).

*Club membership figures are separate, possibly overlapping breakdowns rather than components of a single total 'fitness club/gym' and 'team-sport club' do not sum to 'club membership (any sport)', since a person could belong to both, or to a club type not itemised here.

Continuing recent trends suggests a decline in most individual activities, except cycling, while all three club-membership measures show potential growth. However, this recovery is uneven: team-sport club membership shows the strongest increase (4.7% → 7.1%), whereas fitness/gym membership remains broadly flat or declines (10.0% → 9.4%). The wide scenario ranges for team sports and active travel reflect their strong disruption during COVID, so these long-term projections should be interpreted as directional rather than precise forecasts.

Forecasts of activity levels for each London borough

Applying the propensity model to each borough's projected demographic composition produces a 2025 borough-level forecast ranging from 51.3% in Barking and Dagenham to 75.4% in Wandsworth.

The roughly 24-percentage-point gap between the most and least active boroughs is expected to persist throughout the projection period. The non-overlapping confidence intervals suggest that this difference is

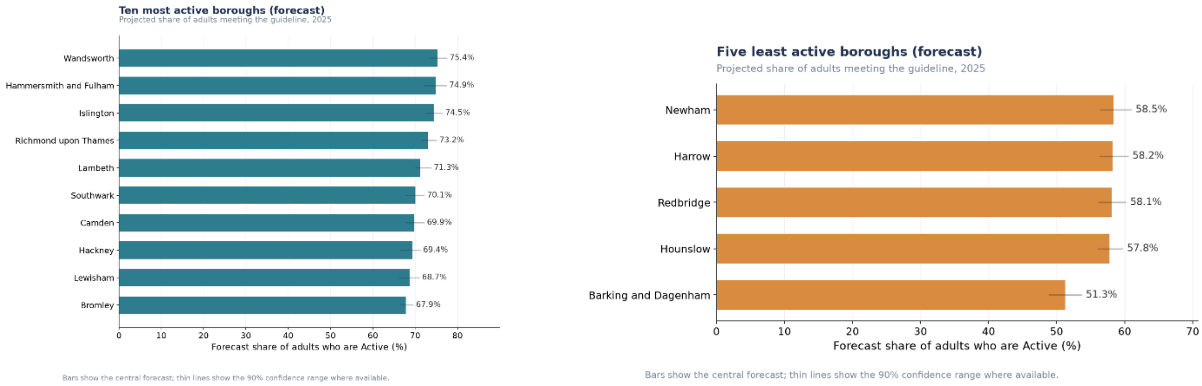

Figure 4.3: Ten most active boroughs & five least active boroughs, projected share of adults meeting the guideline, 2025.

unlikely to be due to sampling variation alone. In addition, all 32 boroughs show a negative composition effect, ranging from close to zero to around -0.9 percentage points, indicating that projected demographic changes are expected to place some downward pressure on activity levels across London, although the size of this effect varies between boroughs.

Key factors that may reduce free time and affect participation

Free-time constraint questions were asked once, in the 2022–23 wave only, of approximately a third of respondents. This section is therefore necessarily a cross-sectional snapshot rather than a ten-year forecast.

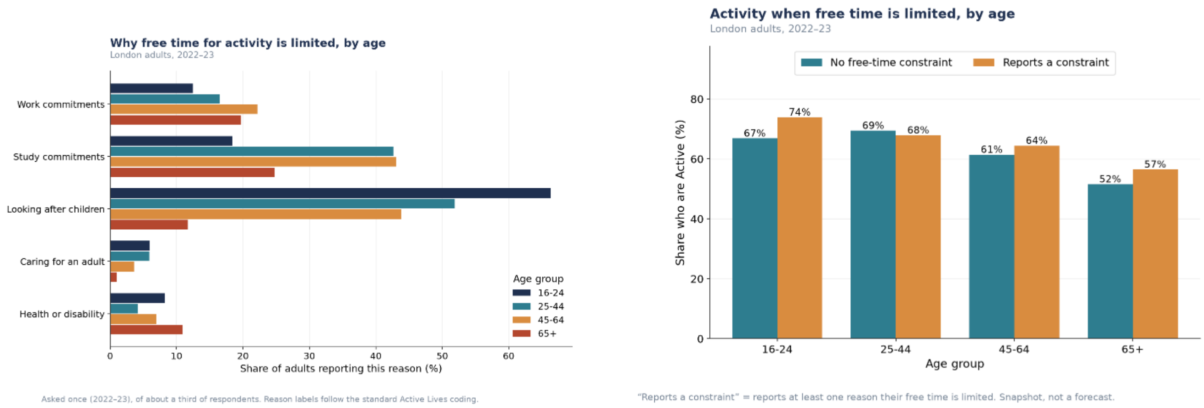

Figure 4.4: Reasons free time for activity is limited & Activity rate by whether free time is constrained, (London adults, 2022–23).

The results show that work, study and childcare are the main reported constraints on free time, particularly among working-age adults, while caring responsibilities and health or disability are less frequently reported. However, reporting any free-time constraint was not associated with lower overall activity; constrained respondents were slightly more likely to be Active. This should be interpreted as a descriptive association rather than a causal relationship, as different constraints may have opposing effects. In particular, long working hours show a clearer negative association with physical activity, while activities such as childcare or active travel may themselves involve physical activity.

4.3.2 P2 - Demographic-Specific Insights

Forecasted participation levels by demographic group

The forecast shows clear differences in physical activity across demographic groups, with socio-economic position showing the widest gap: around 73% activity among the Higher group compared with 50% in the Lower group (23 percentage points). This is followed by the age gap of around 20 points (70% among 16–24 vs 50% among 65+), the disability gap of 19 points, and the ethnicity gap of 13 points. Gender shows the smallest difference, at around 4 points (66% male vs 62% female). The 65+ group is

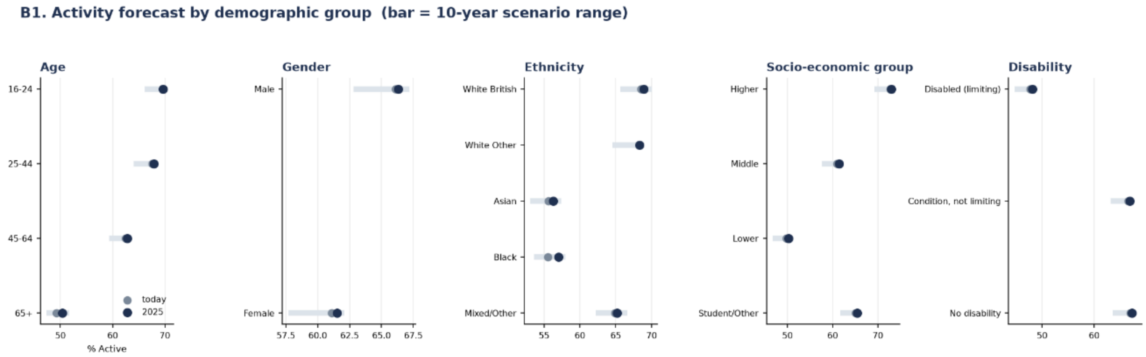

Figure 4.5: Activity forecast by demographic group; bar = 3-year forecast range, dot = today's rate.

particularly important, as it is both the least active group and shows little projected change (around +1 percentage point), meaning London's ageing population may create a demographic headwind to overall activity levels.

Types of sport or physical activity each group is most likely to engage in

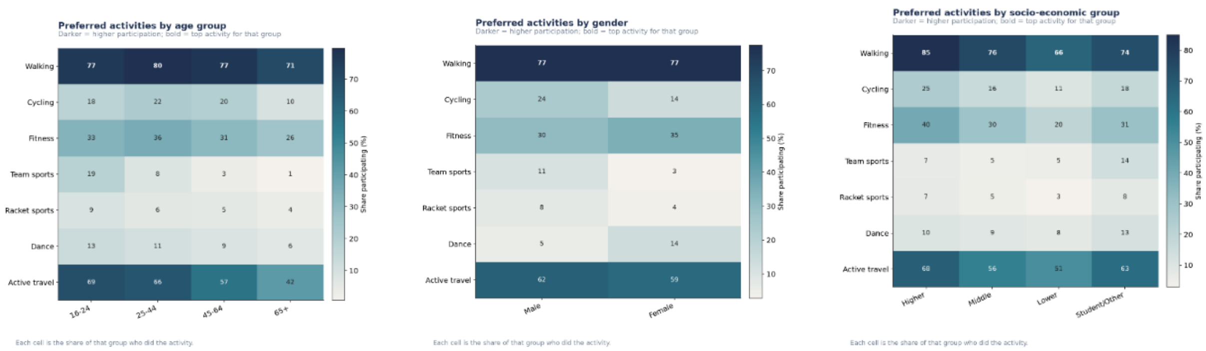

Figure 4.6: Which activities each group does most; bold marks the top activity for that group.

Walking is the most common activity across all demographic groups, although participation is lowest among adults with a limiting disability (64%). The strongest age gradient occurs in team sports, falling sharply from 19% among 16–24-year-olds to just 1% among those aged 65+, while active travel, the second most common activity also declines from 69% to 42%. Gender differences vary by activity: men participate more in cycling, team and racket sports, while women lead in fitness and particularly dance (14% vs 5%), with walking equal at 77%. Ethnic differences are also activity-specific, with Black respondents showing the highest participation in both dance (16%) and team sports (10%), while Asian respondents have the lowest fitness (26%) and active travel (54%) rates.

Socio-economic position shows particularly large differences in facility-dependent activities, with fitness at 40% in the Higher group versus 20% in the Lower group and cycling at 25% versus 11%. However, team sports are highest among the Student/Other group (14%), while dance varies relatively little across socio-economic groups. Similarly, adults with a limiting disability have the lowest participation across all seven activities, with particularly large gaps in fitness, racket sports and active travel, but almost no gap in dance. Overall, dance stands out as the exception: unlike most other activities, it shows relatively small socio-economic and disability differences, suggesting that activities requiring greater access to facilities, equipment or organised provision are more strongly associated with demographic inequalities.

Participation within each London borough

Data note: the ask was for participation in specific activities within each borough. However, the details for activity-type × borough was not available at reportable cell sizes. Thus, the provided plot is broken down by age-group participation within each borough. Borough-by-demographic cells are small: a single borough-by-age cell contains around six respondents in a given wave. All eight waves were therefore pooled and thin cells suppressed.

The 65+ group is the least active in all 32 boroughs, but the youngest group is not always the most active. In 11 boroughs, 25–44 matches or exceeds 16–24, most notably in Borough 012 (Hackney), where activity rises from 64% (16–24) to 76% (25–44). The age gap also varies considerably between boroughs, from 29 points in Borough 027 (Richmond upon Thames) (84% vs 55%) to just 11 points in Borough 014 (Haringey) (65% vs 54%). Borough 032 (Wandsworth) has the highest 65+ rate (59%), while 002 (Barking and Dagenham) has the lowest (36%), showing that borough-level differences can reinforce the effect of age.

Likelihood of participation among people with health conditions

Self-rated health is available from 2018–19 onwards (only 5 years data with 2 years of covid); the health-condition measures apply to people with a limiting disability. Results are descriptive, not forecasts.

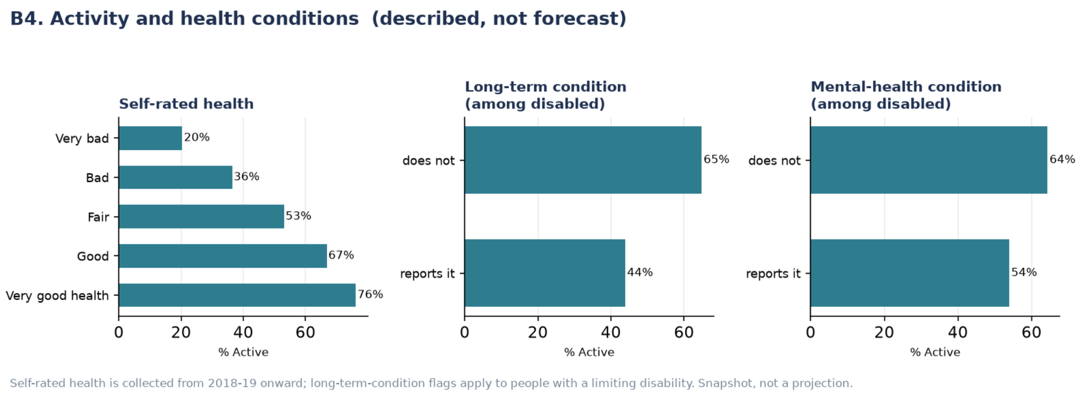

Figure 4.7: Activity by self-rated health, long-term condition and mental-health condition

Activity shows a strong and consistent health gradient, falling from 76% among those reporting very good health to 20% among those reporting very bad health. Among people with a limiting disability, those reporting a long-term condition are 21 percentage points less active than those without one (44% vs 65%), while the gap for mental-health conditions is smaller at 10 points (54% vs 64%). These results show a clear association between poorer health conditions and lower physical activity, although they do not establish causation.

4.3.3 P3 - Indoor vs Outdoor Activity

The expected balance of indoor versus outdoor physical activity

The projection extends only to 2025 and is based on a ∼30% subsample, so it should be treated as indicative rather than a full ten-year forecast.

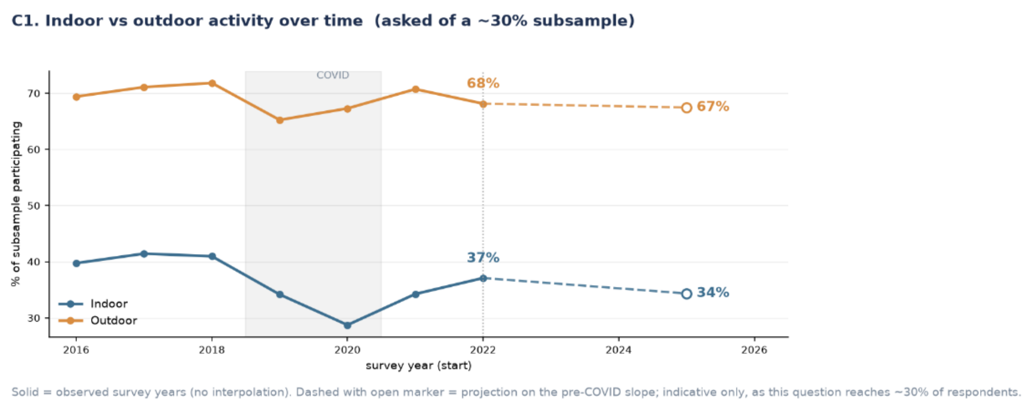

Figure 4.8: Indoor vs outdoor activity over time, observed 2016–2022 and projected to 2025.

Outdoor activity remains substantially more common than indoor activity, at around 65–72% compared with 29–42% over the observed period. Both declined during COVID, but indoor participation fell more sharply, from 41% in 2018 to 29% in 2020, and had only partly recovered to 37% by 2022. The indicative projection suggests indoor activity may fall further to 34% by 2025, while outdoor activity remains broadly stable at 67%, widening the gap between the two. However, this projection has a wide uncertainty range because it is based on the smaller subsample, so the figures should be interpreted directionally rather than as precise forecasts.

Differences in indoor and outdoor activity preferences among people with varying activity levels

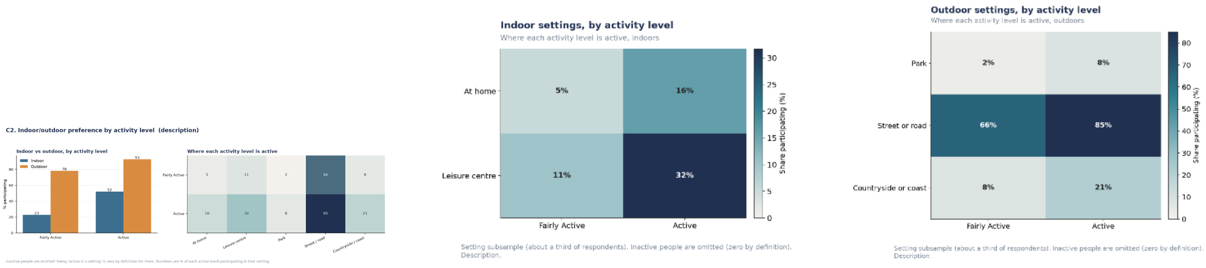

Figure 4.9: Indoor/outdoor preference by activity level, and where each activity level is active.

Indoor activity differs much more by activity level than outdoor activity: Active respondents are more than twice as likely to participate indoors (52% vs 23%), while outdoor participation is already high (93% vs 78%). Street/road activity dominates outdoors (66% vs 85%) and changes least, whereas leisure-centre use shows the largest absolute increase (11%→32%), followed by at-home activity (5%→16%). Park and countryside/coast use also increase (2%→8% and 8%→21%). Overall, Fairly Active people mainly rely on accessible outdoor activity, while Active people are more likely to add indoor and a wider range of activity settings to their routine.

Relationships between inner vs outer London locations, activity levels and socio-economic factors

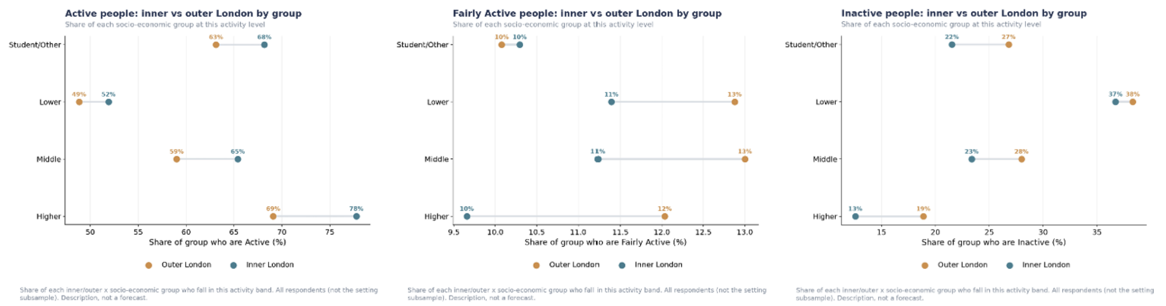

Figure 4.10: Inner vs outer London × socio-economic group, indoor and outdoor activity.

Inner London generally has higher participation than Outer London across all socio-economic groups, but the gap is much smaller for the Lower group (∼2–3 points) and wider for the Higher group, reaching around 10 points for outdoor activity. However, socio-economic differences are still stronger than location. Higher-income respondents in Outer London (40% indoor; 73% outdoor) remain more active than Lower-income respondents in Inner London (22% indoor; 56% outdoor). Overall, where people live matters, but socio-economic position appears to have the bigger influence.

4.3.4 P4 - Volunteering in Sport

Data Note: Volunteering cannot be reliably forecast because the survey question base changed across the years and demographics explain only around 15% of borough-level variation. It is therefore reported descriptively rather than forecast at borough level.

Current volunteering patterns across demographic groups

Volunteering is highest among younger adults, with 16–24-year-olds at 25%, compared with 15% among 65+. Men also volunteer more than women (21% vs 15%), while there is a clear socio-economic gradient (19% Higher vs 12% Lower). In contrast, volunteering varies much less by ethnicity and disability, ranging from 15–20% across ethnic groups and 15–19% by disability status. Overall, age, gender and socio-economic background show stronger differences than ethnicity or disability.

Volunteer role patterns across demographic groups

Organising/administration is the most consistent volunteer role across demographic groups, particularly among middle- and higher-income groups and most ethnicities. Younger adults stand out for more hands-on roles, with 16–24s highest in stewarding (10.1%) and coaching (8.4%). There is also a clear

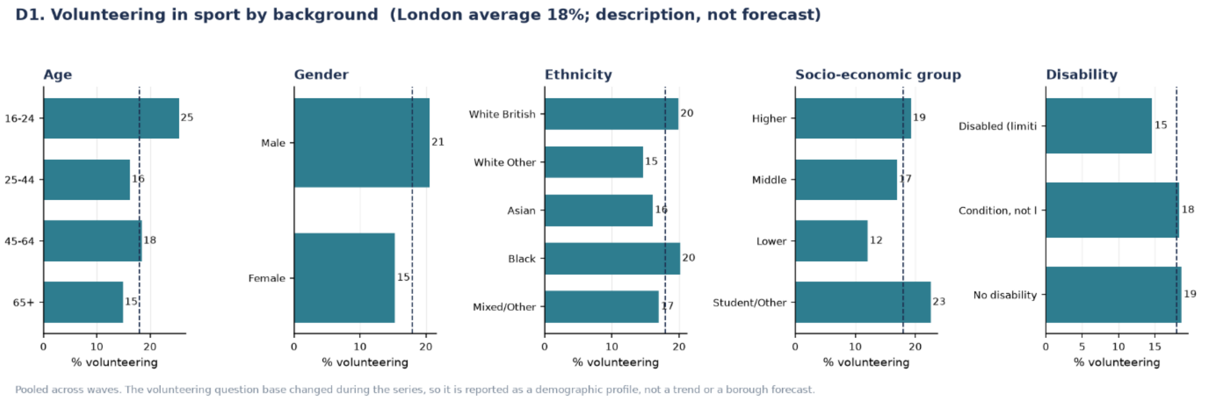

Figure 4.11: Volunteering in sport by background, pooled across waves. Described, not forecast.

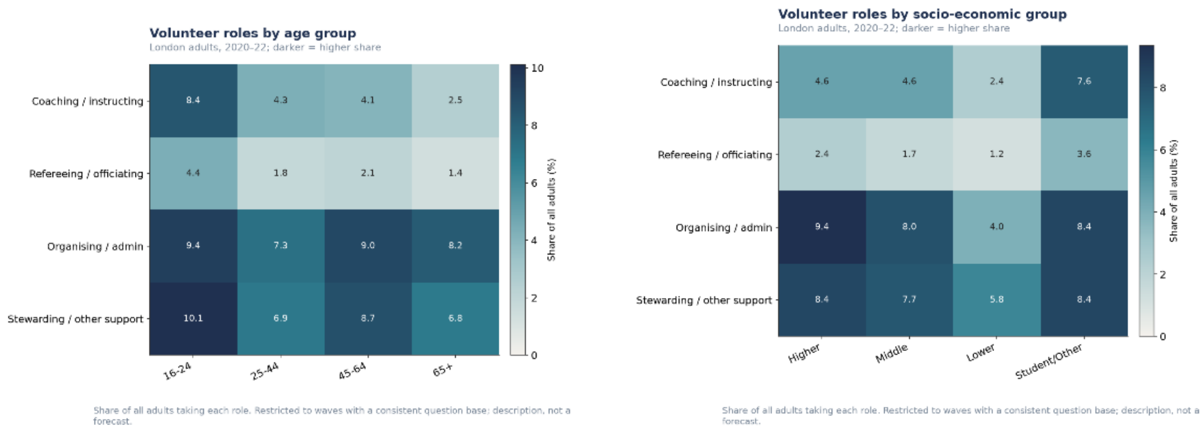

Figure 4.12: Volunteer roles by demographic group, waves 2020–22

gender difference: men are more represented in coaching, officiating and stewarding, while women are more concentrated in organising and support. Lower socio-economic and some disability groups also show relatively higher reliance on stewarding/support roles.

4.5 Critical Discussion

The results provide a consistent picture of physical activity across London. Overall activity is relatively stable, but this stability reflects two opposing effects. An ageing population creates a demographic headwind, while activity levels within demographic groups have generally improved. The estimated -0.37 percentage-point composition effect is partly offset by a $+0.69$ percentage-point behavioural effect. This suggests that maintaining current activity levels may become more challenging as London's population changes.

The demographic results also show clear differences in participation between groups. Socio-economic group shows the widest overall gap, with age and disability also showing substantial differences. The 65+ group is particularly important because it is both the least active and the group showing the smallest improvement over time. These differences are also seen at borough level, where the age gradient remains present across all 32 boroughs. This suggests that demographic characteristics and where people live can work together to shape participation.

These findings have several practical implications for London Sport and local authorities. A single intervention is unlikely to address all of the differences observed. Walking and active travel provide an important low-barrier foundation, while activities such as fitness, cycling and structured sport show larger socio-economic and disability gaps. Improving access to facilities, clubs and suitable local infrastructure could therefore help address some of these differences. The relatively even pattern in volunteering also suggests that volunteering opportunities could provide another way of engaging people from a wider range of backgrounds.

The forecasts should, however, be treated as indicative estimates rather than precise predictions. The Active Lives Survey is repeated cross-sectional data, meaning that the analysis captures changes in the population rather than following the same individuals over time. The forecasting period is also based on a limited number of clean annual observations, with COVID-19 affecting several activity measures. Population projections introduce further uncertainty, while behavioural projections assume that historical patterns provide a reasonable indication of future change. Some outcomes, including volunteering and health-related measures, were therefore reported descriptively rather than forecast.

Overall, the findings suggest that future activity in London will be shaped by both changes in the population and the opportunities available to people. Demographic ageing is difficult to change directly, but the behavioural and place-related factors identified in this analysis provide areas where policy and investment can make a difference. The forecasts are therefore most useful for identifying which groups and areas may need greater attention and where interventions could have the greatest potential impact, rather than as exact predictions of London's future activity levels.

Chapter 5

Conclusion

5.1 Summary of Contributions

This project addressed a gap in how the Active Lives Survey has typically been used: existing research describes historical participation but rarely forecasts future change. Using eight harmonised waves of the Active Lives Adult Survey (2015–16 to 2022–23) across London's 32 boroughs, the project developed a decision support framework that separates the relationship between individual demographic characteristics and activity level from projected changes in future population composition, avoiding the need to fit a conventional time series model to only 256 borough-year observations.

Key contributions include:

1. Harmonisation of eight structurally inconsistent survey waves (292–601 variables each) into a single dataset of 135,497 respondents with standardised ONS borough codes.
2. A propensity-based forecasting framework that decomposes forecasted change into a *composition effect* (demographic shift) and a *behaviour effect* (change within demographic groups), the project's most policy-relevant finding, showing that London's apparently stable overall activity rate masks an ageing-population headwind being offset by genuine behavioural improvement.
3. A four-model comparison (ordinal logistic regression, LightGBM, CatBoost with SHAP, K-Modes clustering, and Bayesian hierarchical modelling) under consistent time-based validation, with LightGBM selected as the final forecasting model (AUC 0.707, ∼16 minute runtime).
4. A composition–place decomposition showing borough identity explains a further 40 percentage points of between-borough variation beyond demographics alone (48% → 88%).
5. Application of the framework across four pillars — overall participation, demographic differences, indoor/outdoor balance, and volunteering — forecasting only where the data supported it and reporting descriptively elsewhere.

5.2 Evaluation Against Aims and Objectives

All five objectives set out in Chapter 1 were met. Harmonisation (Objective 1) is documented in Chapter 3. The propensity model was developed and benchmarked (Objective 2): the ordinal logistic regression offered near-identical performance to LightGBM (AUC 0.704 vs 0.707) with fully transparent coefficients, but the fuller comparison in Chapter 4 — weighing runtime, non-linear interactions, and integration with forecasting — favoured LightGBM as the production model, a reasonable trade-off given it retains feature-importance interpretability while handling the broader activity-type dataset better.

Composition versus place effects (Objective 3) were established via the composition–place decomposition and corroborated independently by the Bayesian random-effects model and the CatBoost/SHAP residual analysis. The forecasting framework itself (Objective 4) is the project's central methodological contribution, directly addressing the eight-wave data limitation that ruled out conventional time series methods. Finally, the framework was applied across all four analytical pillars (Objective 5), with volunteering and indoor/outdoor activity correctly handled descriptively rather than forecast, given their more limited question coverage.

The project therefore achieved its aim: an interpretable, borough-level forecasting framework was built and applied, producing results, notably the composition/behaviour decomposition and the persistent $\sim$24 point gap between the most and least active boroughs, directly usable by a non-technical stakeholder such as London Sport.

5.3 Limitations and Future Work

The survey's repeated cross-sectional design means the analysis captures population-level rather than within-individual change, and COVID-19 disruption reduced usable years for several sub-analyses (health-linked participation, indoor/outdoor split). Borough-level differences are not fully explained by demographic and survey data alone; incorporating external data such as facility density, green space, and cycling infrastructure is a natural next step to help distinguish environmental access from persistent local behaviour differences.

Two further directions stand out. First, the Bayesian hierarchical model showed promise for quantifying borough-level forecast uncertainty but was constrained by long runtimes ($\sim$96 minutes) and limited posterior sampling; a more efficient implementation could move it from a corroborating role to a primary source of uncertainty estimates. Second, as further survey waves become available, volunteering and indoor/outdoor measures such as currently descriptive due to changing question bases and smaller subsamples, could be brought into the forecasting framework once a longer, more consistent time series exists.

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Appendix A

Supplementary Methods and Validation

Concepts, equations and results supporting the methodology and validation chapters
This appendix documents the components of the analysis that are referenced in, or underpin, the main methodology and results but are not set out there in full. Each section states the concept, the relevant equation where applicable, and the result obtained on the real data. Together they complete the white-box account of how the forecasts were produced and validated.

A.1 External validation of the activity bands

The three-band activity outcome (Inactive 0–29, Fairly Active 30–149, Active 150+ weekly moderate-equivalent minutes) was derived directly from the minutes measure rather than taken from a survey-supplied grouping. Deriving it independently permits a direct external check against Sport England's own published classification of the same underlying measure.
Result: the derived classification reproduced Sport England's published grouping with 100% agreement in every one of the eight survey waves. Because this single check exercises the file merge, the recoding and the outcome definition simultaneously against a reference the analysis did not control, it is the foundational validation on which all downstream results rest.

A.2 Non-disclosure as an informative category

The optional demographic questions were declined by some respondents. Rather than dropping or imputing these cases, non-disclosure was retained as an explicit category, because disclosure behaviour is itself strongly predictive of activity.

Disclosure behaviour	Share of sample	% meeting the guideline
Disclosed all optional items	88.1%	65.7%
Withheld one item	11.0%	48.0%
Withheld two or more items	1.0%	28.2%

This is a 37-percentage-point gradient — wider than the effect of age, ethnicity or socio-economic group. Whatever non-disclosure captures (survey engagement, language, institutional trust), it predicts inactivity more strongly than any demographic variable in the data. Imputing or dropping these respondents would overstate activity in every subgroup, so retaining "Not disclosed" as a level is both more honest and more accurate.

A.3 Data-quality audit: four corrected defects

Systematic cross-wave screening — comparing each variable's coverage and prevalence wave by wave and flagging abrupt shifts — uncovered four defects, each of which would have produced confident but wrong results, and none of which was visible by casual inspection.

Defect	What was wrong	Artefact if missed	Correction
Club-membership base	Asked of sport players most years, but of everyone in the final year	Membership appears to collapse 36% → 16%	Rebased to all adults; incomparable early waves excluded
Volunteer-role base	Asked of all until 2019–20, then of volunteers only	Coaching appears to rise fivefold, 5% → 26%	Rebased to all adults; roles restricted to consistent waves
Health variable	Absent in the first three waves	Becomes a disguised indicator of which wave a respondent came from	Excluded from all models; used only for description
Implausible activity	146 respondents report more weekly minutes than a week contains	Distorts group averages	Winsorised at a defensible ceiling

These checks are a permanent part of the pipeline: a future year's data is screened the same way automatically, and any new structural break is flagged rather than silently absorbed.

A.4 Composition versus place: the decomposition of borough variation

Before forecasting, it must be settled whether differences between boroughs are driven by who lives there (composition) or by the place itself, since only the first responds to population change. Let r^b be borough b's observed activity rate and $\hat{r}^b$ its rate predicted from demographics alone. The share of between-borough variation explained by composition is the coefficient of determination across boroughs:

$$R^2_{\text{comp}} = 1 - \frac{\sum_b (r^b - \hat{r}^b)^2}{\sum_b (r^b - \bar{r})^2}$$

Three models were compared on the held-out wave, and the demographic-only model was additionally evaluated leave-one-borough-out (each borough predicted by a model trained without it):

Model	Share of between-borough spread reproduced
Demographics only	48%
Borough identity only	86%
Demographics and borough together	88%
Demographics only, on boroughs never seen in training	47%

Result: roughly half of the difference between boroughs is compositional and half is not. The final row is decisive — the demographic model predicts a borough it has never encountered (47%) almost as well as one it trained on (48%), which is precisely the property that licenses using demographic projection to forecast populations that do not yet exist. Place effects do not offset composition but amplify it: the demographic model ranks boroughs almost perfectly (correlation 0.90) yet understates the size of the gaps, so disadvantaged boroughs are penalised twice — once by demography and once by environment.

A.5 The provision gradient

Repeating the composition-versus-place decomposition across fifteen behaviours produced the project's clearest strategic finding: the more a behaviour depends on facilities and provision, the less of its geography demographics explain.

The ordering is not arbitrary. Walking requires only a pavement and tracks the population closely; cycling requires lanes and storage; park and racket sports require facilities; volunteering requires organisations to volunteer for. The strategic implication is direct: the behaviours least predictable from demographics are precisely those most open to influence through investment. Cycling, the behaviour with the widest borough gap in the dataset, is roughly 71% attributable to place rather than population.

Follows the people (composition-led)		Follows the place (provision-led)	
Walking	50%	Volunteering	15%
Overall activity	48%	Park use	18%
Fitness activities	45%	Team-sport club	25%
Outdoor activity	45%	Cycling	29%
Indoor activity	44%	Leisure-centre use	31%

A.6 Benchmarks and the skill score

A model's quality is meaningful only against a baseline. Two families of benchmark were used. For the classifier, the floor is a prevalence predictor that assigns the base rate to everyone (AUC 0.5 by construction); single-feature models give a stronger floor.

Benchmark	AUC	Interpretation
Prevalence (base rate for everyone)	0.500	The absolute floor
Age band alone	0.572	One predictor
Ethnicity alone	0.568	One predictor
Socio-economic group alone	0.619	Strongest single feature
Full propensity model	0.704	Clears any single feature by 0.085

For the forecast, the benchmarks are computed per borough: persistence (each borough held at its last value) and drift (each borough continuing its own trend). Model quality is then reported as a skill score — the fractional reduction in error relative to a benchmark:

$$\text{Skill} = 1 - \frac{\text{MAE}_{\text{model}}}{\text{MAE}_{\text{benchmark}}}$$

Skill > 0 means the method beats the benchmark; 0 means no better than trivial; < 0 means worse. Reporting skill rather than raw error makes "the model is good" an explicit, quantified comparison. The forecasting framework achieved skill of +0.20 against persistence and +0.31 against drift.

A.7 Interval calibration

A point forecast is only half of a responsible forecast; the stated uncertainty must also be truthful. A 90% interval should contain the actual value about 90% of the time. Coverage is defined as the share of back-test cases whose actual value fell inside the predicted band:

$$\text{Coverage} = \frac{1}{N} \sum_i \mathbb{1} \left[lo_i \le y_i^{actual} \le hi_i \right]$$

Measured on the back-test, the original intervals covered the actual value only 53% of the time — substantially over-confident. The cause was that the bootstrap captured only sampling uncertainty (which respondents were surveyed) and omitted process uncertainty (the genuine year-to-year variation of the world). The published bands were therefore missing an entire source of uncertainty, not merely too small.

The correction is principled, not cosmetic. The process standard deviation was measured empirically from the back-test's own one-step forecast errors (3.44 percentage points) and added to the sampling standard deviation in quadrature:

$$\sigma_{\text{total}} = \sqrt{\sigma^2_{\text{samp}} + \sigma^2_{\text{proc}}} \qquad 90\% \text{ interval} = \hat{r} \pm 1.645\, \sigma_{\text{total}}$$

Interval basis	Measured 90% coverage
Sampling only (original)	53% — over-confident
Sampling + measured process variance (calibrated)	94%

Note on honesty: the widening was fixed by the measured process variance, not tuned to hit a target, and the resulting coverage (94%, marginally conservative) was reported as it fell out. A result of exactly 90% would in fact be more suspicious. The central forecasts are unchanged by calibration — only the band widths.

A.8 Decomposition of the uncertainty range

A complete uncertainty range reflects three separate sources: the survey happening to sample different people, the model's own coefficients being imperfect, and the genuine year-to-year variation of the process. The bootstrap refits the model on each resample, so it already captures the second source; its size was confirmed by isolating it.

Source	Contribution to the 90% half-width (pp)	Role
Year-to-year process variation	5.66	Dominant — the piece calibration adds
Survey sampling of people	0.44	Small
Model coefficient variability	0.35	Smallest — but genuinely present

The model's own coefficient uncertainty is therefore not missing from the range; it is captured by the bootstrap and is the smallest of the three contributors. The dominant source is year-to-year process variation — precisely what the calibration in Section G was built to include.

A.9 Sensitivity to the shrinkage weight (λ)

The population projection blends each borough's own trend with the London-wide trend, in a balance set by a shrinkage weight λ, fixed at a conservative midpoint of 0.5:

$$\widetilde{m}_{b,c} = \lambda \cdot m_{b,c} + (1-\lambda) \cdot m_c^{London}, \qquad \lambda = 0.5$$

To confirm this choice does not drive the result, the entire forecast was re-run across the full range of λ, from 0 (each borough's own trend only) to 1 (the London trend only), both for the London headline and for every borough separately.

Result: the choice of λ is immaterial. Across the whole range the London forecast moves by just 0.04 percentage points, and even the single most λ-sensitive borough, Waltham Forest, moves only 1.3 points; every other borough moves less. $\lambda = 0.5$ is therefore a safe, conservative default rather than an influential free parameter.

A.10 Corroboration: forecasting for the right reason

A forecast can beat a benchmark by accident. The corroboration check tests whether it beats it for the reason claimed. Because the method forecasts a borough by projecting its demographic composition, it should forecast best on boroughs whose activity is demographically explained, and worst on boroughs driven by place effects it does not model. Writing $|e^b|$ for a borough's composition residual (its place strength, Section D) and MAE^b for its mean back-test error, the mechanism predicts a positive rank correlation:

$$\rho = \text{Spearman}\left(|e^b|, \text{MAE}^b\right) > 0$$

Result: the rank correlation is +0.74 (Pearson +0.65) — a strong positive relationship. The boroughs the framework forecasts worst are precisely those where place dominates (for example Wandsworth and Camden), and the ones it forecasts best are those where demographics explain activity (for example Harrow and Merton). The forecast therefore succeeds through the demographic mechanism it claims, not by coincidence. With 32 boroughs and a small number of origins this is reported as strong corroborating evidence rather than proof.

A.11 Validation summary

The three headline validation results, established on real data, are:

Validation result	Value	What it establishes
Forecast skill vs naïve	+0.20	The forecast is accurate
Calibrated interval coverage	94%	The uncertainty is honest
Corroboration correlation	+0.74	It is accurate for the right reason

Taken together these establish that the forecast is accurate, that its stated uncertainty is truthful, and that its accuracy arises from the demographic mechanism the framework claims. The remaining limitations — principally the three clean back-test origins available from an eight-wave, two-pandemic-year series — are constraints of the data rather than the method: the analysis is more rigorous than the data can fully reward, and the framework strengthens automatically as further survey years arrive.